Decent Coaching Classes

VIII CBSE

Mind Your Math VIII

ANSWER KEY **Marks: 30**

SECTION A (1 mark each)

1. (C) trinomial 2. (B) $3x + \sqrt{x} + 4$ 3. (A) $3xy$ 4. (C) $4x(x + 3)$ 5. (D) 5 6. (B) 12

Hints: 2: $\sqrt{x} = x^{1/2}$, the power is not a whole number. 4: HCF of $4x^2$ and $12x$ is $4x$. 5: $3x = 15$, so $x = 5$. 6: $x = 8 \times 3/2 = 12$.

SECTION B (2 marks each)

7. (a) $3a$ – **monomial**; (b) $2x + y$ – **binomial**; (c) $5x - 2y + 3xy$ – **trinomial**; (d) $a^2 + a - 5$ – **trinomial**. (½ mark each)
8. $(3a - 2ab - 6) - (4a + 5ab - 7) = 3a - 2ab - 6 - 4a - 5ab + 7 =$ **$-a - 7ab + 1$**
9. HCF of $12x^2y$ and $18xy^2$ is $6xy$. $12x^2y - 18xy^2 =$ **$6xy(2x - 3y)$**
10. $5x - 7 = 3x + 9$, so $5x - 3x = 9 + 7$, so $2x = 16$, so **$x = 8$**

SECTION C (3 marks each)

11. $(3a + 5b)(5a + 7b) = 3a \times 5a + 3a \times 7b + 5b \times 5a + 5b \times 7b$
    $= 15a^2 + 21ab + 25ab + 35b^2 =$ **$15a^2 + 46ab + 35b^2$**

    OR $(4x + y)(7x - 6y) = 28x^2 - 24xy + 7xy - 6y^2 =$ **$28x^2 - 17xy - 6y^2$**
12. $6x(x - 3) = 6x^2 - 18x$; $x(2 - 5x) = 2x - 5x^2$.
    Sum $= 6x^2 - 18x + 2x - 5x^2 + 3x^2 =$ **$4x^2 - 16x$** $[= 4x(x - 4)]$
13. (i) $a^2 + 10a + 25 = a^2 + 2(a)(5) + 5^2 =$ **$(a + 5)^2$**
    (ii) $16x^2 - 9y^2 = (4x)^2 - (3y)^2 =$ **$(4x - 3y)(4x + 3y)$**
    (iii) $3x^2 - 12 = 3(x^2 - 4) =$ **$3(x - 2)(x + 2)$**
14. $2(x + 3) - 5 = 3x - 4$, so $2x + 6 - 5 = 3x - 4$, so $2x + 1 = 3x - 4$, so $1 + 4 = 3x - 2x$, so **$x = 5$**.
    Check: LHS $= 2(5 + 3) - 5 = 11$; RHS $= 3(5) - 4 = 11$. LHS = RHS ✓

    OR $x/2 + x/3 = 10$. Multiply by 6: $3x + 2x = 60$, so $5x = 60$, so **$x = 12$**. Check: $12/2 + 12/3 = 6 + 4 = 10$ ✓

SECTION D (1 mark each)

15. (i) Perimeter $= 2[(2x + 3) + (x + 4)] = 2(3x + 7) =$ **$(6x + 14)$ m**
    (ii) $6x + 14 = 50$, so $6x = 36$, so **$x = 6$**
    (iii) Area $= (2x + 3)(x + 4) = 2x^2 + 8x + 3x + 12 =$ **$2x^2 + 11x + 12$** sq m
    (iv) Length $= 2(6) + 3 =$ **15 m**; breadth $= 6 + 4 =$ **10 m**. (Check: area = 150 sq m; $2(36) + 66 + 12 = 150$ ✓)